

21. (a) $n=10$ $S_{xx}=7918.16$
 $\bar{x}=26.49$ $S_{xy}=7717.13$
 $\bar{y}=27.2$

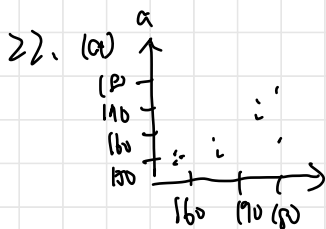
$$b_1 = \frac{S_{xy}}{S_{xx}} = 0.97461$$

$$b_0 = \bar{y} - b_1 \bar{x} = 1.38257$$

$$\hat{y} = 1.38257 + 0.97461x$$

$\therefore$ 每增加 1 cm, actual measurement 平均增加 0.97461 cm

(b) $r^2 = \frac{SSR}{SST} = \frac{7521.195}{7567.89} = 0.9938$



(b) $y=x \therefore \text{slope} = 1$

(c) $n=5$

$$\bar{x}=166$$

$$S_{xx}=568$$

$$\bar{y}=166.375$$

$$S_{xy}=488$$

$$b_1 = \frac{S_{xy}}{S_{xx}} = 0.82394$$

$$b_0 = 29.60035$$

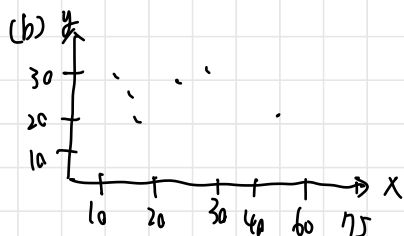
$$\hat{y} = 29.60035 + 0.82394x$$

$$(a) \hat{y} = 29.60035 + 0.82394X$$

$$x=157, \quad y=158.958$$

10, (a) $x = \text{income}$: independent variable

$y = \text{BMI}$: dependent variable



(b) $n=6$

$$\bar{x} = 40.833$$

$$\bar{y} = 27$$

$$b1 = \frac{S_{xy}}{S_{xx}} = -0.14755$$

$$b0 = \bar{y} - b1\bar{x} = 27 - (-0.14755)(40.833) = 32.9144$$

$$\hat{y} = 32.9144 - 0.14755x$$

$\therefore$ Income 每增加 1000, BMI 会下降 0.14755

(C) $SST = 72.200$

$$SSR = 59.442$$

$$SSE = 12.778$$

Source	df	SS	MS	F
regression	1	59.442	59.442	18.8
error	4	12.778	3.195	
total	5	72.200		

income 会 82.31% 的 BMI variation

11- (a) $n=5$

$$S_{xx} = 96080$$

$$S_{xy} = 64386$$

$$b_1 = \frac{S_{xy}}{S_{xx}} = 0.67013$$

$$b_0 = \bar{y} - b_1 \bar{x} = 195.902$$

$\therefore \hat{y} = 195.902 + 0.67013x$ 每多100本, 時間增加約 67.013 month

(b)



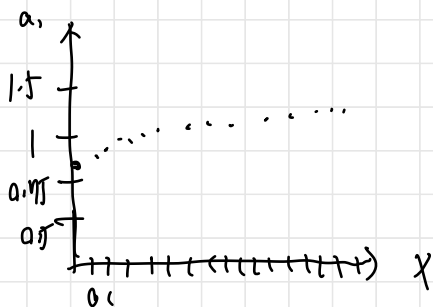
正向關係, 書的數量越多, 所需時間越長

$\therefore$ good fit

(c)

Source	df	SS	MS	F
regression	1	43146.93	43146.93	69.582
error	3	1860.21	620.09	
Total	4	45007.2		

9-



$$b_1 = \frac{S_{xy}}{S_{xx}} = 0.02309$$

$$b_0 = \bar{y} - b_1 \bar{x} = 1.04125 - (0.02309)(7.5) = 0.88809$$

$$\hat{y} = 0.02309x + 0.8881$$

→ 每年新增道路平均长度 0.02309

$$H_0: \beta_1 = 0 \quad SSE = 0.005132$$

$$H_a: \beta_1 \neq 0 \quad MSE = 0.000366$$

$$SE(b_1) = \sqrt{\frac{0.000366}{349}} = 0.001038$$

$$t_{0.025, 14} < |22.235| \therefore \text{reject } H_0$$

$$C. SST = 0.186315$$

$$SSR = 0.181243$$

$$SSE = 0.005132$$

Source	df	SS	MS	F
regression	1	0.181243	0.181243	494.393
error	14	0.005132	0.000366	
Total	15	0.186315		

$$F = \frac{MSR}{MSE} = 494.364$$

$$\alpha = 0.05 \Rightarrow F_{0.05, 1, 14} \approx 4.6 \therefore \text{reject } H_0$$

d. $r^2 = 0.9725$, about 97.25% of variation in roadway miles is explained by the linear relationship with years.

$$10. a. b_0 = 93.617 \quad \hat{y} = 93.617 - 1.251X$$

$$b_1 = -1.251 \rightarrow \text{每年新增1英里, 再次 assessed 的概率下降 1.25\%}$$

$$b. H_0: \beta_1 = 0$$

$$H_0: \beta_1 = 0$$

$$t_{0.025, 5} = 2.571, |t| > 2.571, \therefore \text{reject } H_0$$

$$c. 1.25 \pm 2.571(0.120) \Rightarrow (-1.559, -0.943)$$

我們有 95% 的信心區間 group median age 每增加 1, percentage arrested avg 會下降 約 0.943 ~ 1.559 %

d. $R^2 = 0.9564 \Rightarrow$ 約 95.64% 的 percentage arrested variation 可藉由 group median age linear model 解釋